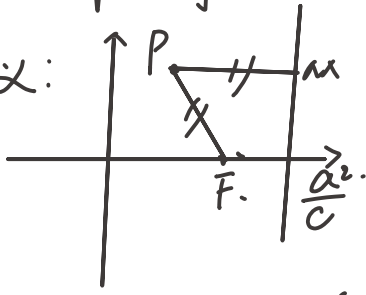


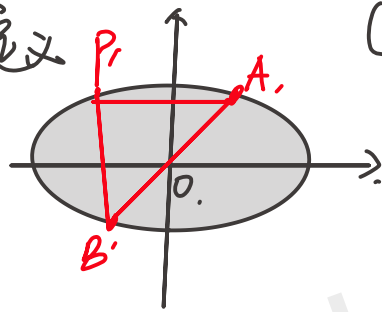
第6讲 圆锥曲线核心知识体系

高考前最后一遍核心全面梳理笔记:

一、六大定义:

1. 第一定义: 椭圆: $PF_1 + PF_2 = 2a$, 双: $|PF_1 - PF_2| = 2a$.2. 第二定义:  $\frac{PF}{PM} = \frac{c}{a} \Rightarrow \begin{cases} 0 & \text{圆} \\ (0,1) & \text{椭圆} \\ 1 & \text{抛物} \\ (1,+\infty) & \text{双} \end{cases}$

3. 第三定义



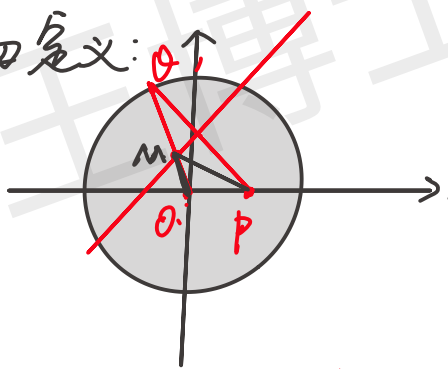
(点差法)

$$k_{PA} \cdot k_{PB} = -\frac{b^2}{a^2} \text{ 椭圆} = e^2 - 1 \text{ (-x轴)}$$

$$= \frac{b^2}{a^2} \text{ 双}$$

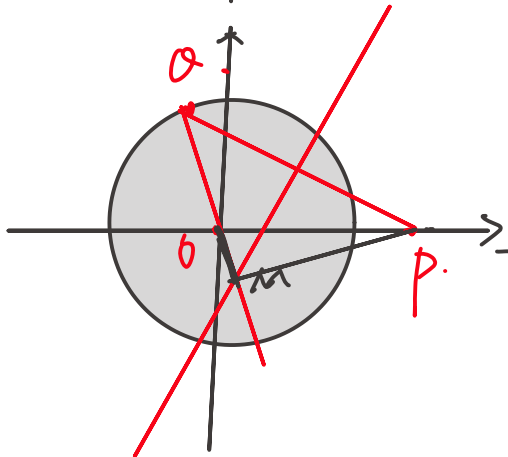
$$k_{PA} \cdot k_{PB} = \frac{1}{e^2 - 1} \text{ (y轴)}$$

4. 第四定义:



$$MP + MO = MO + MQ = r = \text{const.}$$

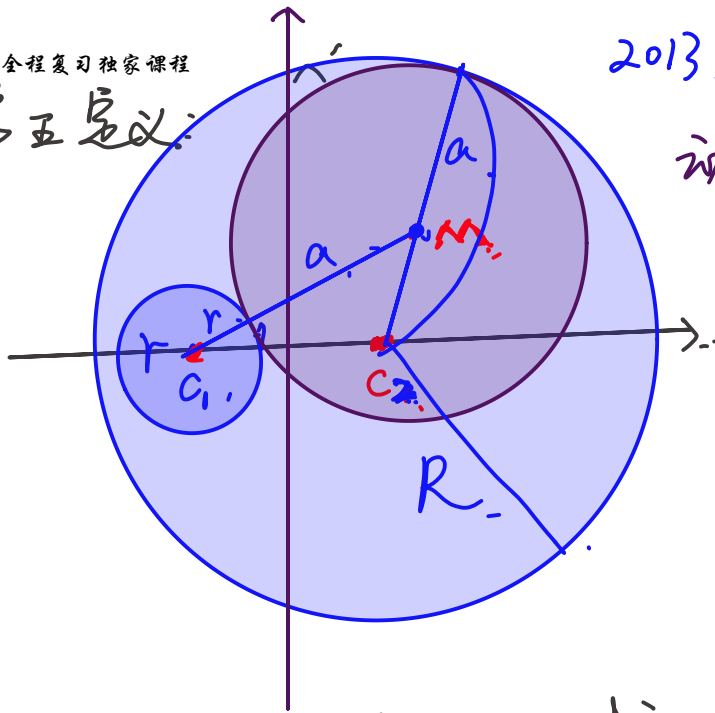
M 轨迹: 椭圆.



$$MP - MO = MQ - MO = \text{const.}$$

M 轨迹: 双曲线.

5. 第五定义:



2013 新课标 第二阶段

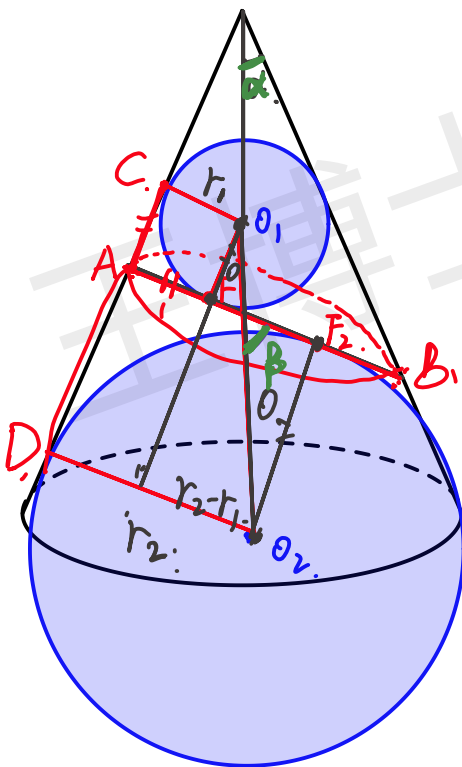
动圆 M .

设 $m = a$.

$$MC_1 + MC_2 = a + r + R - a$$

椭圆

6. 第六定义: 丹德林双球



$$AF_2 = AD, AC = AF_1$$

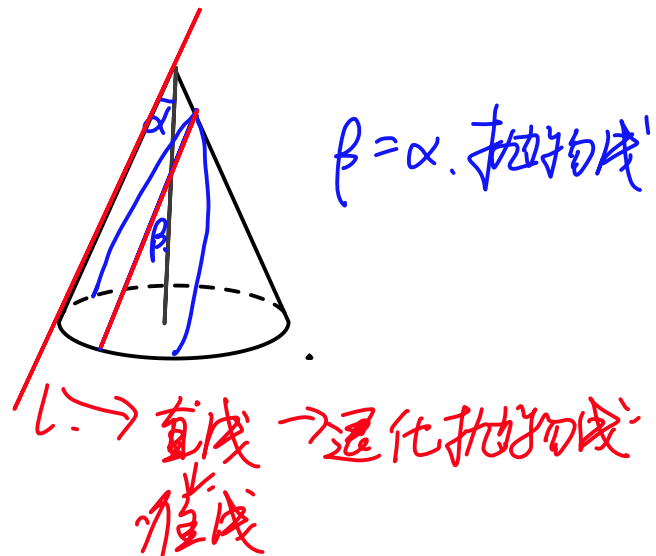
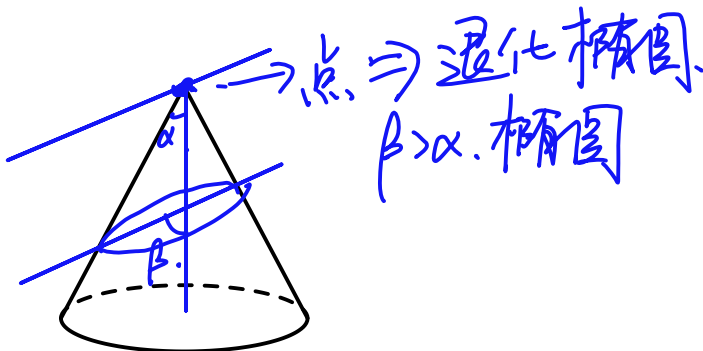
$$CD = 2a, \quad CD = AF_1 + AF_2 = 2a.$$

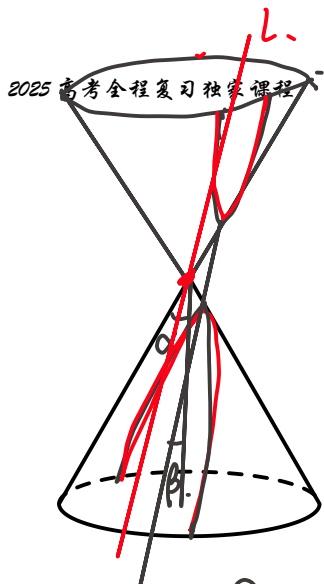
$\beta > \alpha$. 椭圆.

$\beta = \alpha$. 抛物线.

$\beta < \alpha$. 双曲线.

$$e = \frac{\cos \beta}{\cos \alpha}$$

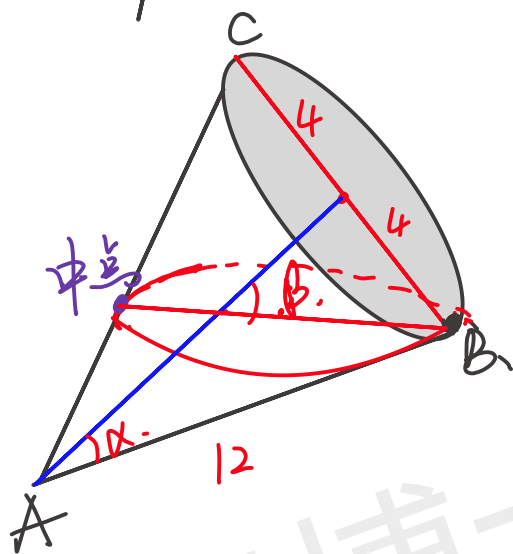




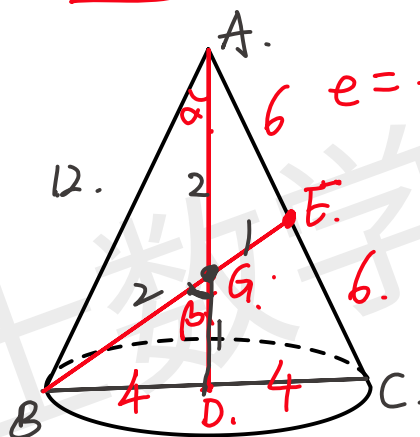
$\alpha > \beta$. 双曲线.



$\rightarrow$ 2条直线 $\Rightarrow$ 退化双曲线.
渐近线.



$e = ?$



$$e = \frac{\cos \beta}{\cos \alpha}$$

G 重心.

$$\cos \alpha = \frac{\sqrt{144-16}}{12} = \frac{2}{3}\sqrt{2}.$$

$$\underline{BE \text{ 中线}} = \frac{1}{2} \sqrt{2(AB^2 + BC^2) - AC^2}.$$

$$= \frac{1}{2} \sqrt{2(12^2 + 64) - 144} = 2\sqrt{17}.$$

$$\underline{BG} = \frac{2}{3} \cdot 2\sqrt{17}. \quad \underline{GD} = \frac{1}{3} AD = 2\sqrt{2}.$$

$$\cos \beta = \frac{2\sqrt{2}}{\frac{4}{3}\sqrt{17}}$$

$$e = \frac{\cos \beta}{\cos \alpha}.$$

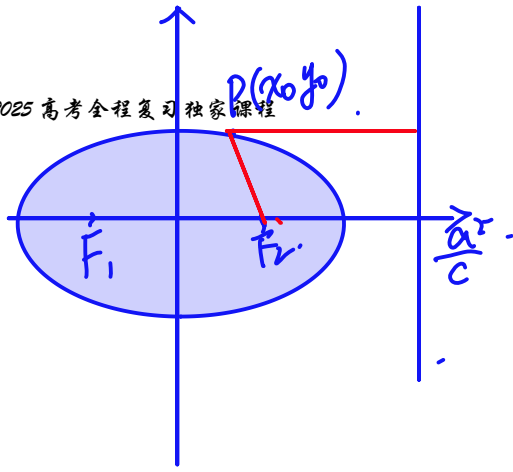
二. 焦半径与焦点三角形.

1. 第一焦半径公式:

椭圆: $\frac{2b^2}{a}$.

双: $\frac{2b^2}{a}$.

抛物线: $2p$.



$$\frac{PF_2}{\frac{a^2}{c} - x_0} = \frac{c}{a} = e$$

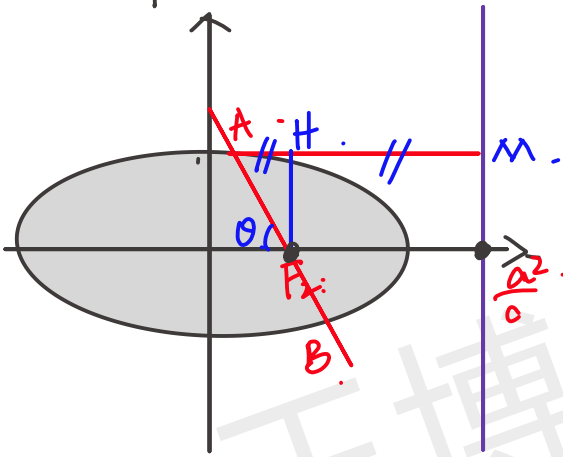
$$PF_2 = a - ex_0$$

$$PF_1 = a + ex_0$$

椭圆: 双: $\left\{ \begin{array}{l} PF_1 = |a + ex_0| \\ PF_2 = |a - ex_0| \end{array} \right.$

抛物线: $PF = x_0 + \frac{p}{2}$

2 第二焦半径公式



$$HM = \frac{a^2}{c} - c = \frac{b^2}{c}$$

$$AH = AF_2 \cos \theta$$

$$\frac{AF_2}{AF_2 \cos \theta + \frac{b^2}{c}} = \frac{c}{a}$$

$$AF_2 = \frac{c}{a} AF_2 \cos \theta + \frac{b^2}{a}$$

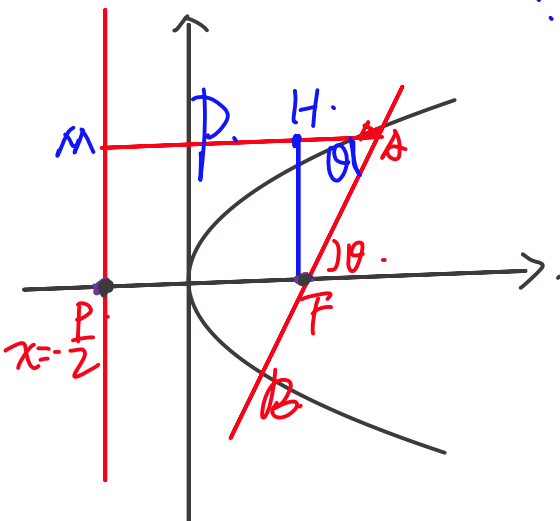
$$AF_2 = \frac{\frac{b^2}{a}}{1 - e \cos \theta}$$

$$BF_2 = \frac{\frac{b^2}{a}}{1 + e \cos \theta}$$

$$AB = \frac{\frac{2b^2}{a}}{1 - e^2 \cos^2 \theta}$$

$$\frac{b^2}{a} \cdot e \cdot \frac{1}{a}$$

椭圆: 双:



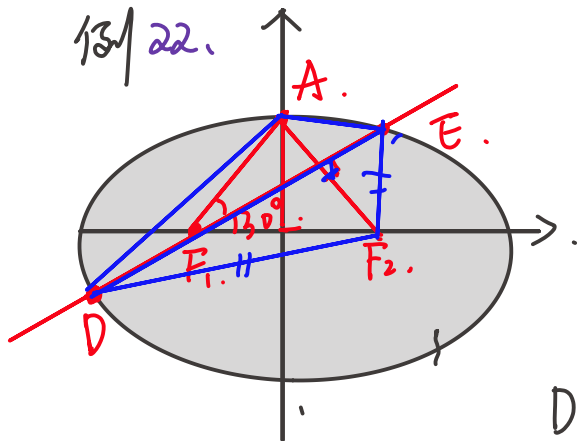
$$AF = AF \cos \theta + p$$

$$AF = \frac{p}{1 - \cos \theta}$$

$$BF = \frac{p}{1 + \cos \theta}$$

$$AB = \frac{2p}{1 - \cos^2 \theta} = \frac{2p}{\sin^2 \theta}$$

13/22.



$$e = \frac{1}{2} \quad |DE| = 6.$$

 $\triangle ADE$ 周长 = 13.

$$a = 2c.$$

$$b = \sqrt{3}c.$$

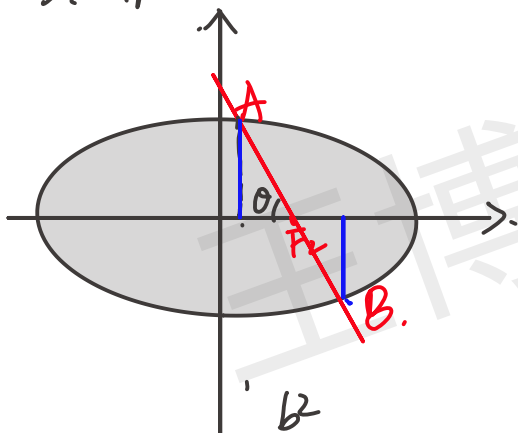
 $\triangle EDF$ 周长 $4a$.

$$DE = \frac{\frac{2b^2}{a}}{1 - \frac{1}{4} \cdot \frac{3}{4}} = 6.$$

$$\frac{c^2}{2c} = \frac{13}{16}.$$

$$c = \frac{13}{8} \quad a = \frac{13}{4} \quad 4a = 13.$$

3. 焦点弦定理.



$$\vec{AF_2} = \lambda \vec{F_2B}.$$

1. 大题: 相似, 消x.

$$\begin{cases} y_A = -\lambda y_B \\ y_A + y_B = \dots \\ y_A y_B = \dots \end{cases}$$

2. 小题: 焦点弦定理.

$$AF_2 = \frac{\frac{b^2}{a}}{1 - e \cos \theta} = \lambda BF_2 = \frac{\lambda \frac{b^2}{a}}{1 + e \cos \theta}.$$

$$\Rightarrow e \cos \theta = \left| \frac{\lambda - 1}{\lambda + 1} \right|$$

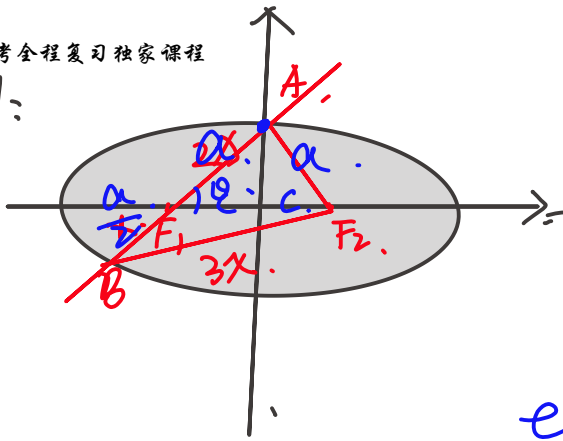
$$\text{由: } \frac{1}{\cos^2 \theta} - \frac{\tan^2 \theta}{k^2} = 1 \quad \text{椭圆.}$$

$$e = \sqrt{1 + k^2} \left| \frac{\lambda - 1}{\lambda + 1} \right| \quad \text{双曲线.}$$

$$\text{抛物线: } \cos \theta = \left| \frac{\lambda - 1}{\lambda + 1} \right|$$

$$\sqrt{1 + k^2} \left| \frac{\lambda - 1}{\lambda + 1} \right| = 1.$$

例:



$$AF_1 = 2BF_1$$

$$AB = BF_2$$

$$\text{求 } e = \underline{\hspace{2cm}}$$

$$x + 3x = 2a$$

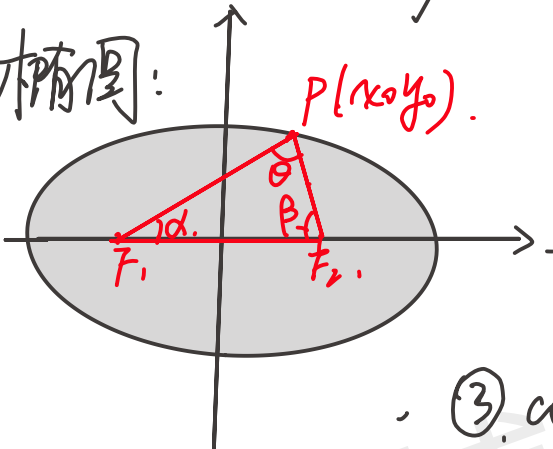
$$x = \frac{a}{2}$$

$$e \cdot \left(\frac{c}{a}\right) = \left| \frac{2-1}{2+1} \right| \Rightarrow e^2 = \frac{1}{3}$$

$$e = \frac{\sqrt{3}}{3}$$

4. 焦点三角形.

椭圆:

① 周长 $2a + 2c$.② 焦点 + 离心率 $e = \frac{c}{a} = \frac{2c}{2a}$.

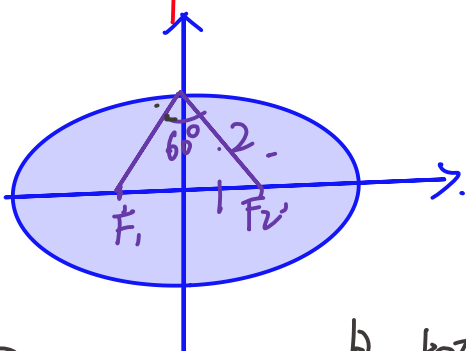
$$e = \frac{F_1F_2}{PF_1 + PF_2} = \frac{\sin(\alpha + \beta)}{\sin \alpha + \sin \beta} \text{ 定又几何法.}$$

$$\textcircled{3} \cos \theta = \frac{(PF_1 + PF_2)^2 - 2PF_1 \cdot PF_2 - 4c^2}{2PF_1 \cdot PF_2}$$

$$\frac{4b^2}{2PF_1 \cdot PF_2} = 1 + \cos \theta$$

$$|PF_1| |PF_2| = \frac{2b^2}{1 + \cos \theta}$$

$$S_{\triangle PF_1F_2} = \frac{1}{2} \cdot \frac{2b^2}{1 + \cos \theta} \cdot \sin \theta = b^2 \cdot \frac{\sin \theta}{1 + \cos \theta} = b^2 \tan \frac{\theta}{2}$$

④ P 在短轴端点, $\angle F_1PF_2$ 最大.例: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 存在 P $\angle F_1PF_2 = 60^\circ$ e 的范围 $[\frac{1}{2}, 1)$ 

$$e = \frac{1}{2}$$

$$PF_1, PF_2 \in [a - c, a + c]$$

⑤ $P(x_0, y_0)$ 是椭圆上一个点.

$$PF_1 = a + ex_0, \quad PF_2 = a - ex_0$$

$$|PF_1||PF_2| = a^2 - e^2 x_0^2, \quad x_0^2 \in [0, a^2]$$

$$= a^2 - \frac{c^2}{a^2} x_0^2 \in [b^2, a^2]$$

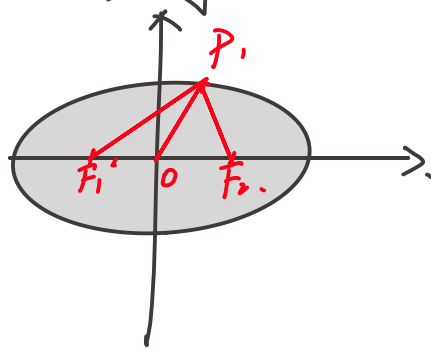
⑥ $P(x_0, y_0)$ 在椭圆上.

$$\overrightarrow{PF_1} \cdot \overrightarrow{PF_2} = PO^2 - \frac{1}{4} \cdot 4c^2$$

$$= PO^2 - c^2.$$

$$PO^2 \in [b^2, a^2]$$

$$\overrightarrow{PF_1} \cdot \overrightarrow{PF_2} \in [b^2 - c^2, b^2]$$



例: P 是 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 上一点. $\overrightarrow{PF_1} \cdot \overrightarrow{PF_2} = c^2$ 也范围

$$b^2 - c^2 \leq c^2 \leq b^2.$$

$$a^2 - 2c^2 \leq c^2 \leq a^2 - c^2.$$

$$\frac{1}{3} \leq e^2 \leq \frac{1}{2}$$

$$e \in \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{2}}{2}\right]$$

⑦ 若 $|PF_1||PF_2| = \lambda$, 求 $S_{\triangle PF_1F_2} = \underline{b\sqrt{\lambda - b^2}}$. $b^2 \tan \frac{\theta}{2}$



$$\lambda = \frac{2b^2}{1 + \cos \theta} \Rightarrow \cos \theta = \frac{2b^2}{\lambda} - 1$$

$$\cos \theta = \frac{\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2}} = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}.$$

$$\underline{\underline{\frac{1}{2} \theta = \tan^{-1} x}}.$$

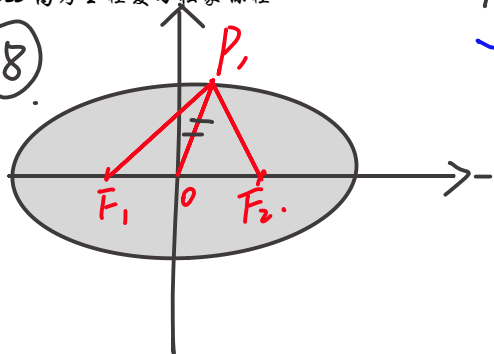
$$\frac{2b^2}{\lambda} - 1 = \frac{1 - x^2}{1 + x^2} = -1 + \frac{2}{1 + x^2}$$

$$x^2 = \frac{\lambda - b^2}{b^2}.$$

$$x = \frac{\sqrt{\lambda - b^2}}{b}$$

$$S_{\triangle PF_1F_2} = b^2 \tan \frac{\theta}{2} = b^2 x = \underline{b\sqrt{\lambda - b^2}}. \quad \star$$

⑧



$$|OP| = \lambda, \quad S_{\triangle PF_1F_2} = \frac{b\sqrt{a^2 - \lambda^2}}{\text{第二阶段}}$$

$$OP = \frac{1}{2} \sqrt{2(PF_1^2 + PF_2^2) - F_1F_2^2} = \lambda$$

$$= \frac{1}{2} \sqrt{2[(PF_1 + PF_2)^2 - 2PF_1 \cdot PF_2] - 4c^2} = \lambda$$

$$= \frac{1}{2} \sqrt{8a^2 - 4PF_1 \cdot PF_2 - 4c^2} = \lambda$$

$$|PF_1| \cdot |PF_2| = 2a^2 - c^2 - \lambda^2$$

$$S_{\triangle PF_1F_2} = b\sqrt{2a^2 - c^2 - \lambda^2} = b\sqrt{a^2 - \lambda^2}$$

$$\textcircled{9} \quad PF_1 = \lambda PF_2, \quad S_{\triangle PF_1F_2} =$$

$$(1+\lambda)PF_2 = 2a$$

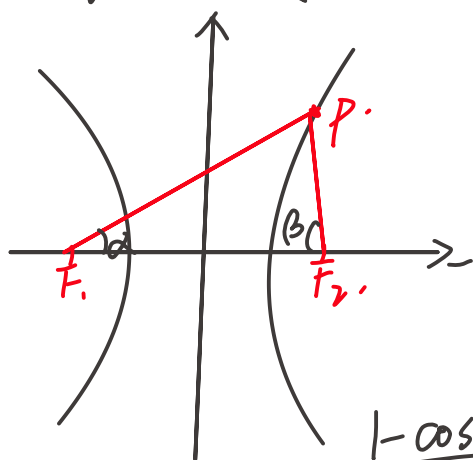
$$PF_2 = \frac{2a}{1+\lambda}$$

$$PF_1 = \frac{2a\lambda}{1+\lambda}$$

$$PF_1 \cdot PF_2 = \frac{4a^2\lambda}{(1+\lambda)^2}$$

$$S_{\triangle PF_1F_2} = b\sqrt{\frac{4a^2\lambda}{(1+\lambda)^2} - b^2}$$

双曲线:



$$\textcircled{1} \quad e = \frac{c}{a} = \frac{2c}{2a} = \frac{F_1F_2}{|PF_1 - PF_2|} = \frac{\sin(\alpha + \beta)}{|\sin\alpha - \sin\beta|}$$

$$\textcircled{2} \quad |PF_1| \cdot |PF_2| = \frac{2b^2}{1 - \cos\theta}$$

$$S_{\triangle PF_1F_2} = b^2 \frac{\sin\theta}{1 - \cos\theta} = b^2 \frac{1}{1 - \cos\theta} = b^2 \tan \frac{\theta}{2}$$

$$\frac{1 - \cos\theta}{\sin\theta} = \frac{\sin\theta}{1 + \cos\theta} = \tan \frac{\theta}{2}$$

$$\textcircled{3}. |PF_1| |PF_2| = \lambda \Rightarrow \lambda = \frac{2b^2}{1 - \cos \theta} \Rightarrow \cos \theta = 1 - \frac{2b^2}{\lambda}$$

$$\cos \theta = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} \Rightarrow \tan \frac{\theta}{2} \Rightarrow S = b^2 / \tan \frac{\theta}{2}$$

$$\textcircled{4} |OP| = \lambda. |OP| = \frac{1}{2} \sqrt{2(PF_1^2 + PF_2^2) - F_1F_2^2} = \lambda \Rightarrow |PF_1 \cdot PF_2| \Rightarrow \textcircled{3}$$

$$\textcircled{5}. PF_1 = \lambda PF_2. PF_2 = \frac{2a}{\lambda - 1} PF_1 = \frac{2a\lambda}{\lambda - 1} |PF_1 \cdot PF_2| \Rightarrow \textcircled{3}$$

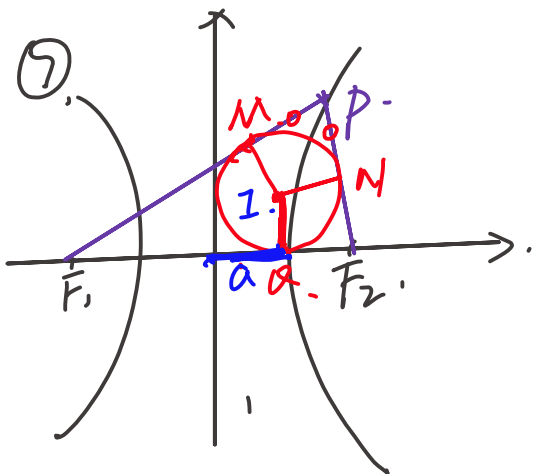
$\textcircled{6} P(x_0, y_0)$ 右支上一点.

$$|PF_1| = |a + ex_0|, |PF_2| = |a - ex_0|$$

$$|PF_1| \in [a + c, +\infty), |PF_2| \in [c - a, +\infty)$$

焦半径, 求 e 范围. 约束条件不够.
从自身找范围.

右支: $x_0 \geq a \Rightarrow$ 范围 (23, 合肥一模16)



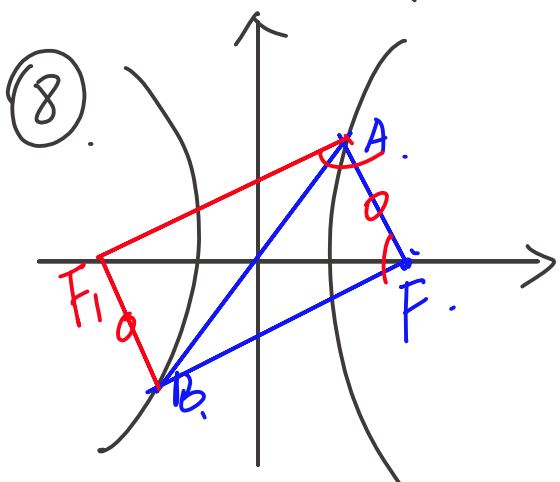
$$PF_1 - PF_2 = 2a.$$

$$MF_1 - MF_2 = 2a.$$

$$F_1O - OF_2 = 2a \Rightarrow F_1O = a + c.$$

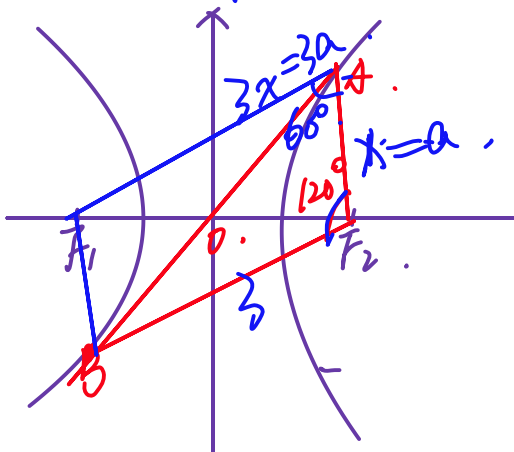
$$F_1O + OF_2 = 2c \Rightarrow F_2O = c - a.$$

$$x_I = a.$$



过原点的直线 { 1. 第三定义
2. 焦点:
补全平行四边形

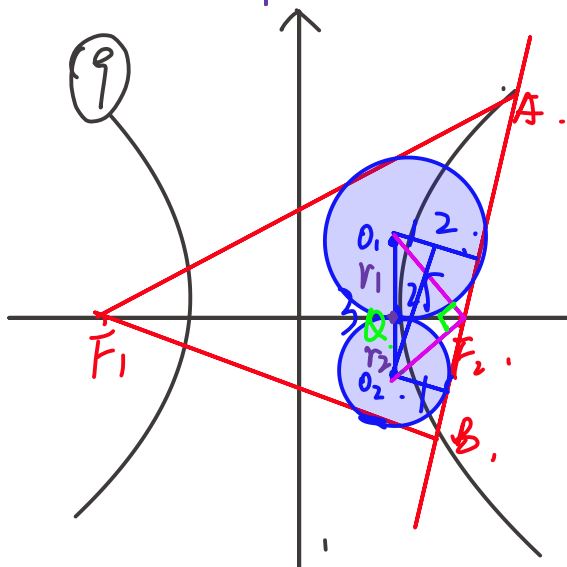
例: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. 右焦点 F . 过原点直线交 A, B .
 $\angle AFB = 120^\circ$. $BF = 3AF$. $e = \frac{\sqrt{7}}{2}$.



$$2x = 2a \\ x = a$$

$$\cos 60^\circ = \frac{9a^2 + a^2 - 4c^2}{6a^2} = \frac{1}{2}$$

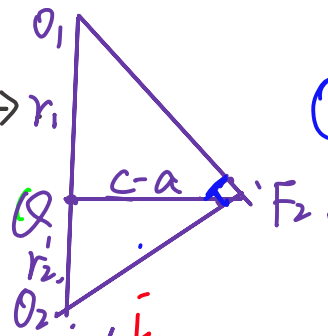
$$\begin{cases} a^2 = 4c^2 \\ e = \frac{\sqrt{7}}{2} \end{cases}$$



① $x_{O1} = x_{O2} = a$

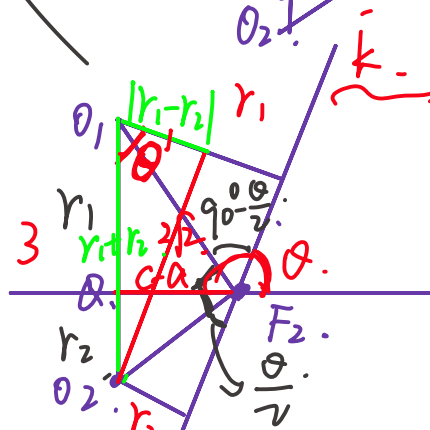
② $\angle O_1 F_2 O_2 = 90^\circ$

③ $r_1 r_2 = (c-a)^2$



AB 是右支

$|k| > \left|\frac{b}{a}\right|$



$$r_1 = (c-a) \cdot \tan\left(90^\circ - \frac{\theta}{2}\right)$$

$$r_2 = (c-a) \cdot \tan \frac{\theta}{2}$$

$$\cos \theta = \frac{|r_1 - r_2|}{|r_1 + r_2|}$$

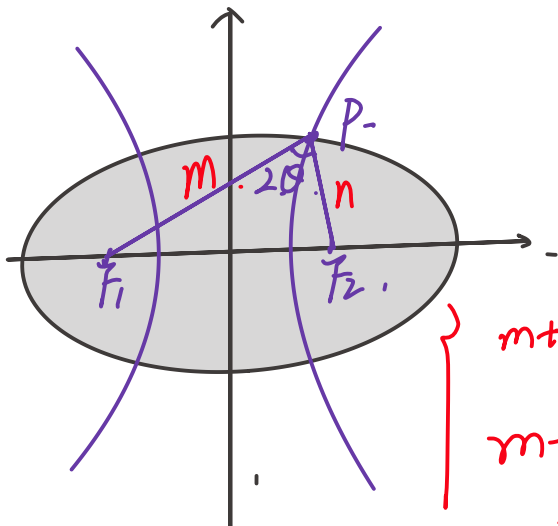
若 $r_1 = 2r_2$. 求 L 的斜率

$$k = \tan \theta = \frac{2\sqrt{2}}{1} = 2\sqrt{2}$$

⑩: 共焦点问题: 椭圆: a_1, e_1 .

双曲线: a_2, e_2 .

求证: $\frac{\sin^2 \theta}{e_1^2} + \frac{\cos^2 \theta}{e_2^2} = 1$



$$\begin{cases} m+n=2a_1 \\ m-n=2a_2 \end{cases} \Rightarrow \begin{cases} m=a_1+a_2 \\ n=a_1-a_2 \end{cases}$$

$$\cos 2\theta = \frac{(a_1+a_2)^2 + (a_1-a_2)^2 - 4c^2}{2(a_1^2 - a_2^2)}$$

$$2(a_1^2 - a_2^2) \cos 2\theta = 2a_1^2 + 2a_2^2 - 4c^2$$

$$a_1^2(1 - \cos 2\theta) + a_2^2(1 + \cos 2\theta) = 2c^2$$

$$a_1^2 \sin^2 \theta + a_2^2 \cos^2 \theta = c^2$$

$$\frac{\sin^2 \theta}{e_1^2} + \frac{\cos^2 \theta}{e_2^2} = 1$$

例: $\angle F_1 P F_2 = 60^\circ$. 椭圆: e_1 . 双: e_2 . 共焦点.

求 $\frac{1}{e_1} + \frac{\sqrt{3}}{e_2}$ 最大值 $2\sqrt{2}$

$$\frac{\frac{1}{4}}{e_1^2} + \frac{\frac{3}{4}}{e_2^2} = 1 \Rightarrow \frac{1}{e_1^2} + \frac{3}{e_2^2} = 4$$

求 $\frac{1}{e_1} + \frac{\sqrt{3}}{e_2}$ max

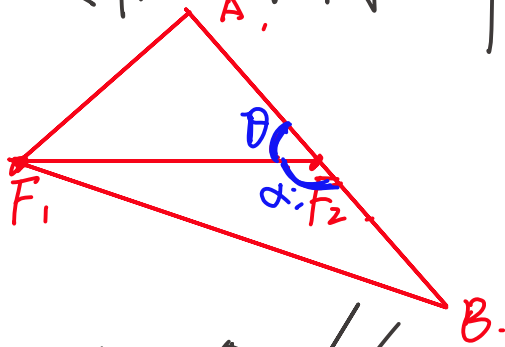
法1: 三角换元.

法2: 柯西不等式

$$\left(\frac{1}{e_1} + \frac{\sqrt{3}}{e_2} \right)^2 \leq (1^2 + 1^2) \left(\frac{1}{e_1^2} + \frac{3}{e_2^2} \right) = 8$$

三. 三线模型在圆锥曲线中命题 (命题人偏爱)

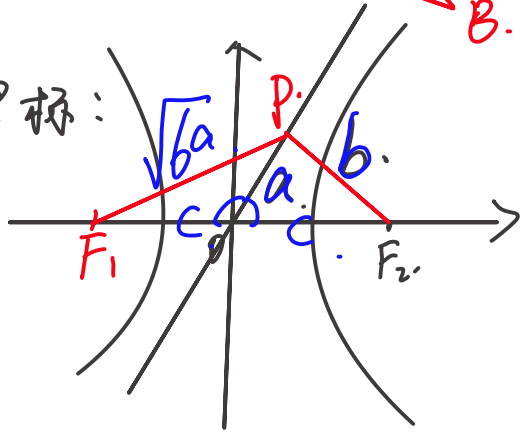
模型:



三线: 通法: 余弦和为0

$$\cos \theta + \cos \alpha = 0$$

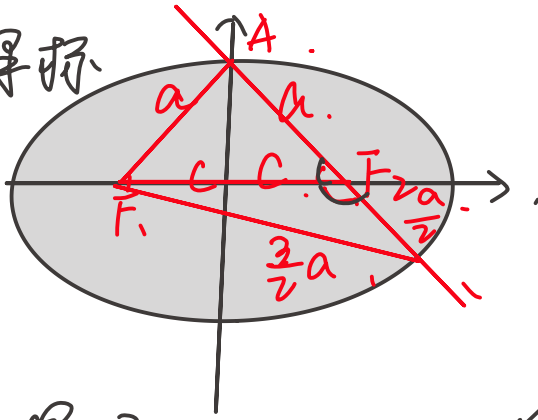
18. 新坐标:



$$PF_1 = \sqrt{6}OP, \text{ 求 } e = \underline{\hspace{2cm}}$$

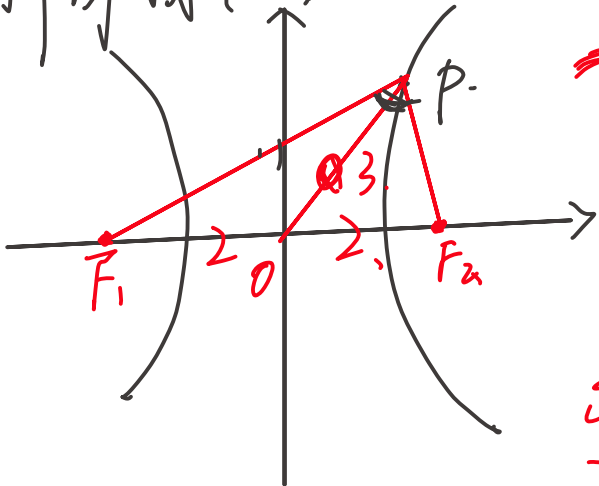
$$\frac{a^2 + c^2 - b^2}{2ac} + \frac{a^2 + c^2 - b^2}{2ac} = 0$$

19. 新坐标



余弦和为0

20. 新坐标:



$$\frac{x^2}{1} - \frac{y^2}{3} = 1, |OP| = 3$$

$$S_{\triangle PF_1F_2} = \underline{\hspace{2cm}}$$

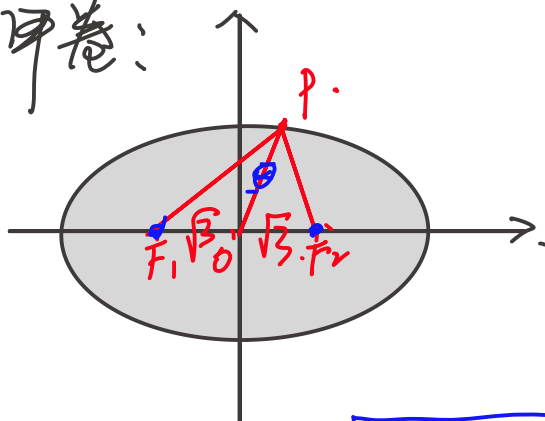
$$3 = \frac{1}{2} \sqrt{2(PF_1^2 + PF_2^2) - 16}$$

$$3 = \frac{1}{2} \sqrt{2(4 + 2PF_1PF_2) - 16}$$

$$PF_1 \cdot PF_2 = \lambda$$

$$S = 6\sqrt{c^2 - b^2}$$

23 甲卷:



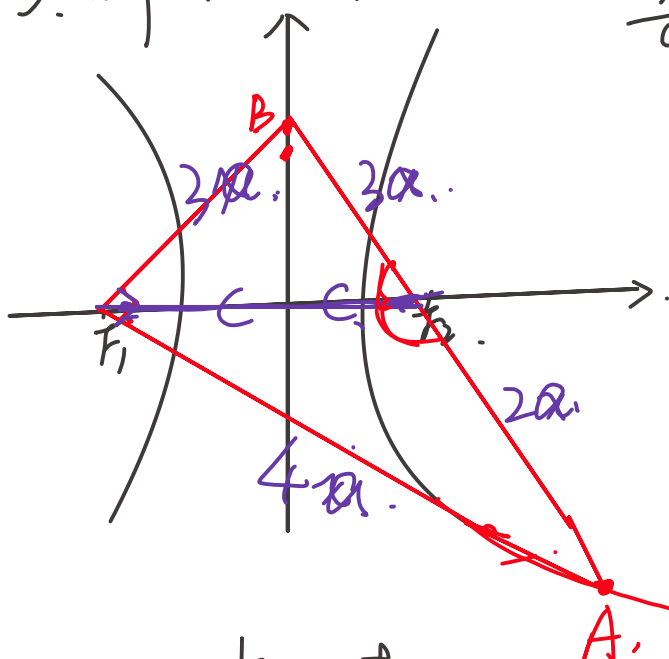
$$\frac{x^2}{9} + \frac{y^2}{6} = 1 \quad \cos \angle F_1 P F_2 = \frac{3}{5}$$

求 $|OP| =$ _____

$$|PF_1| \cdot |PF_2| = \frac{2b^2}{1 + \cos \theta} = \frac{12}{1 + \frac{3}{5}} = \frac{15}{2} \checkmark$$

$$OP = \frac{1}{2} \sqrt{2(36 - 15) - 12} = \frac{\sqrt{30}}{2}$$

23. 新1卷16.



$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\vec{F_1 A} \perp \vec{F_1 B}$$

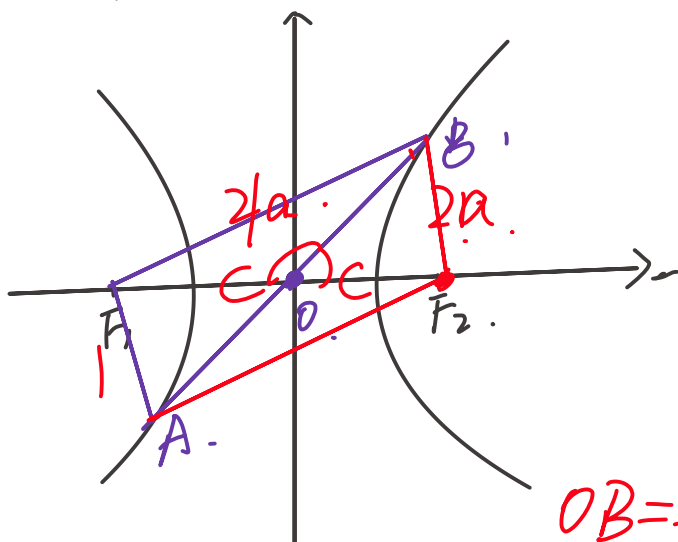
$$\vec{F_2 A} = -\frac{2}{3} \vec{F_2 B} \quad e = \underline{\hspace{2cm}}$$

$$4x - 2x = 2a$$

$$x = a$$

净和为0.

24. 九省联考:



$$F_1 B = 2 F_1 A$$

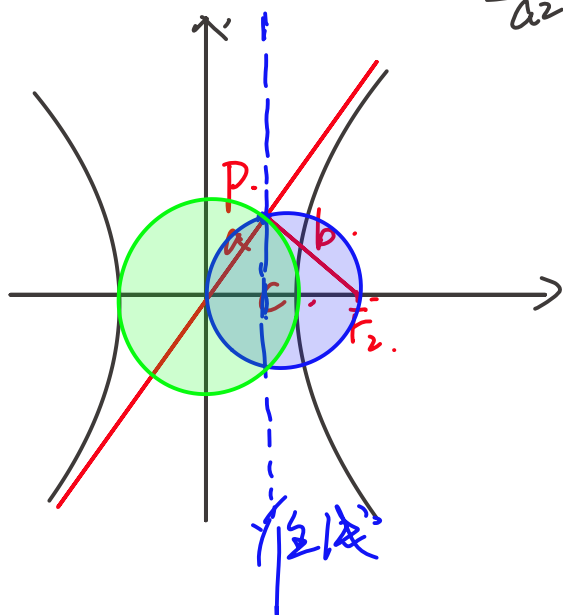
$$\vec{F_2 A} \cdot \vec{F_2 B} = 4a^2 \quad \text{求 } e = \underline{\hspace{2cm}}$$

$$c^2 - OB^2 = 4a^2$$

$$OB^2 = c^2 - 4a^2$$

$$OB = \frac{1}{2} \sqrt{2(16a^2 + 4a^2) - 4c^2} = \sqrt{c^2 - 4a^2}$$

四：渐近线： $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \Rightarrow$
 $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 0$



特征三角形

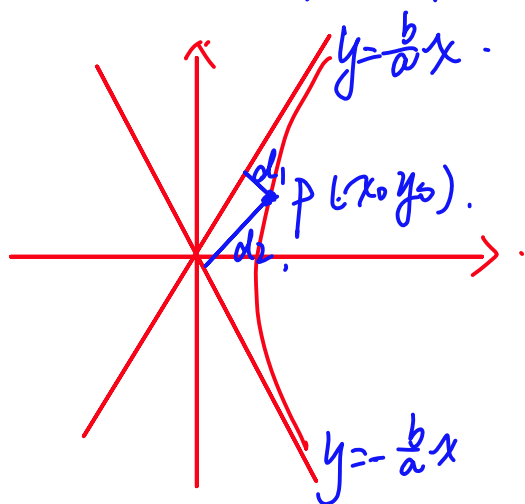
$$x_P = \frac{a^2}{c}, y_P = \frac{ab}{c}$$

$$P\left(\frac{a^2}{c}, \frac{ab}{c}\right)$$

$$P \text{ 在 } \left(x - \frac{c}{2}\right)^2 + y^2 = \frac{c^2}{4} \text{ 上}$$

$$P \text{ 在 } x^2 + y^2 = a^2 \text{ 上}$$

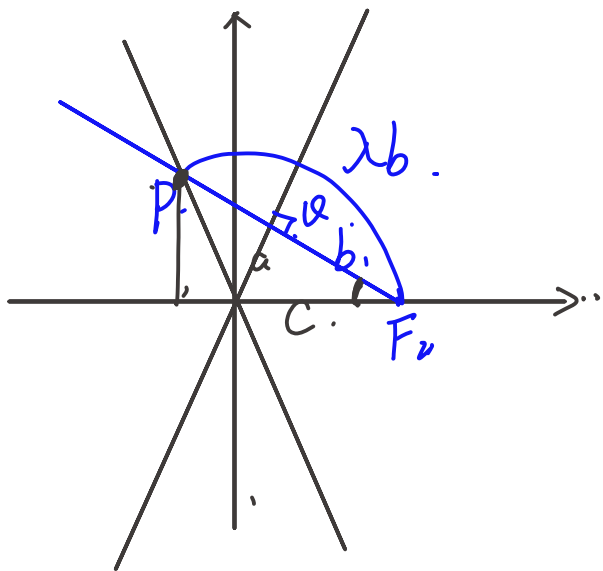
结论：双曲线上的点 P 到两渐近线距离之积为定值 $\frac{a^2 b^2}{c^2}$



$$d_1 = \frac{|y_0 - \frac{b}{a}x_0|}{\sqrt{1 + \frac{b^2}{a^2}}} = |y_0 - \frac{b}{a}x_0| \cdot \frac{a}{c}$$

$$d_2 = \frac{|y_0 + \frac{b}{a}x_0|}{\sqrt{1 + \frac{b^2}{a^2}}} = |y_0 + \frac{b}{a}x_0| \cdot \frac{a}{c}$$

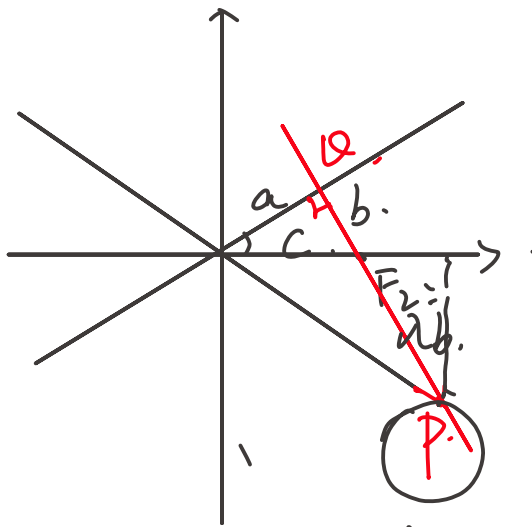
$$d_1 d_2 = \frac{a^2}{c^2} \left| y_0^2 - \frac{b^2}{a^2} x_0^2 \right| = \frac{a^2 b^2}{c^2}$$



$$\vec{F_2 P} = \lambda \vec{F_2 O} \quad e = \sqrt{\frac{2\lambda}{\lambda - 1}}$$

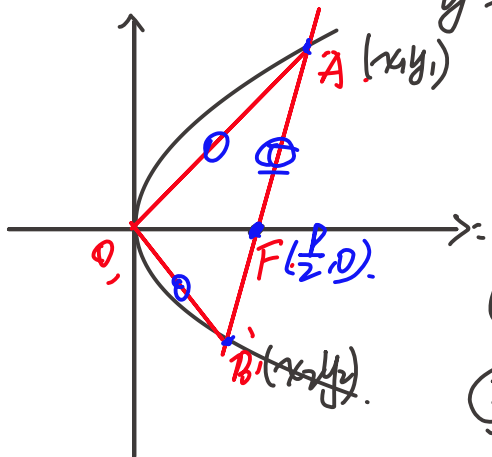
$$x_P = \text{代 } \lambda, y = -\frac{b}{a}x$$

$$y_P =$$



$$\vec{PF_2} = \lambda \vec{F_2Q} \cdot e = \sqrt{\frac{2\lambda}{\lambda+1}}$$

五: 抛物线: $y^2 = 2px$



$$\textcircled{1} \begin{cases} x_1 x_2 = \frac{p^2}{4} \\ y_1 y_2 = -p^2 \end{cases} \quad (AB \text{ 过 } x \text{ 轴上 } (m, 0), x_1 x_2 = m^2)$$

$$\textcircled{2} \vec{OA} \cdot \vec{OB} = x_1 x_2 + y_1 y_2 = -\frac{3}{4} p^2$$

$$\textcircled{3} k_{OA} \cdot k_{OB} = \frac{y_1}{x_1} \cdot \frac{y_2}{x_2} = -4$$

$$\textcircled{4} k_{AB} = \frac{2p}{y_1 + y_2} : k_{AB} = \frac{y_1 - y_2}{x_1 - x_2} = \frac{y_1 - y_2}{\frac{y_1^2}{2p} - \frac{y_2^2}{2p}} = \frac{2p}{y_1 + y_2}$$

$$\textcircled{5} AB: (y_1 + y_2)y = 2px + y_1 y_2 : y = \frac{2p}{y_1 + y_2}(x - x_1) + y_1$$

$$(y_1 + y_2)y = 2px - 2px_1 + y_1^2 + y_1 y_2$$

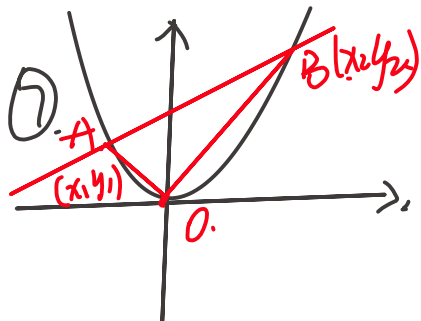
$$(y_1 + y_2)y = 2px + y_1 y_2$$

$$\textcircled{6} k_{OA} = \frac{y_1}{x_1} = \frac{y_1}{\frac{y_1^2}{2p}} = \frac{2p}{y_1}$$

$$k_{OB} = \frac{2p}{y_2}$$

$$k_{AB} = \frac{2p}{y_1 + y_2}$$

$$\frac{1}{k_{OA}} + \frac{1}{k_{OB}} = \frac{1}{k_{AB}}$$

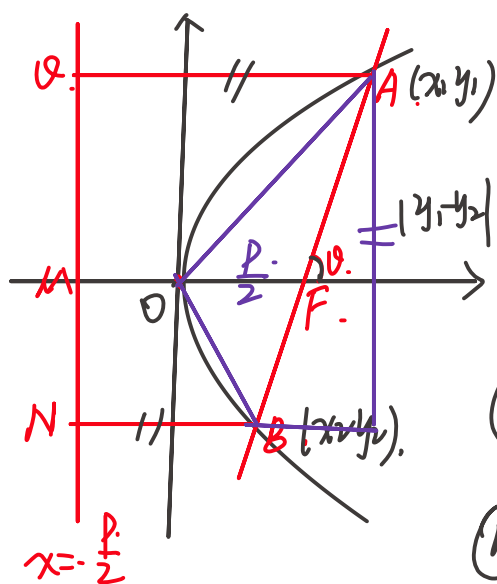


$$k_{AB} = \frac{x_1 + x_2}{2p}$$

$$k_{OA} = \frac{x_1}{2p}$$

$$k_{OB} = \frac{x_2}{2p}$$

$$k_{OA} + k_{OB} = k_{AB}$$



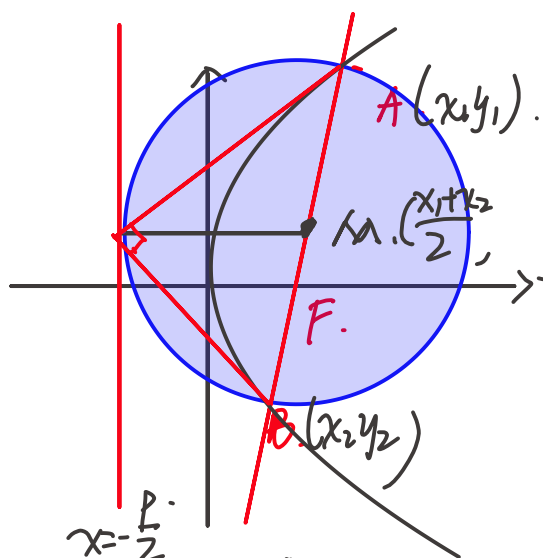
⑧: $AF = x_1 + \frac{p}{2}$ $AB = x_1 + x_2 + p$
 $BF = x_2 + \frac{p}{2}$

⑨: $AF = \frac{p}{1 - \cos \theta}$ $AB = \frac{2p}{1 - \cos^2 \theta} = \frac{2p}{\sin^2 \theta}$
 $BF = \frac{p}{1 + \cos \theta}$

⑩: $\frac{1}{AF} + \frac{1}{BF} = \frac{2}{p}$

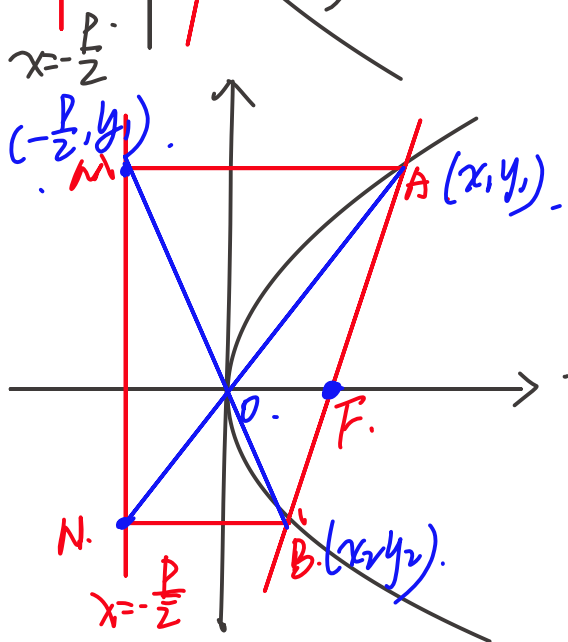
⑪: $S_{\triangle AOB} = \frac{1}{2} \cdot \frac{p}{2} \cdot AB \cdot \sin \theta = \frac{p}{4} \cdot \frac{2p}{\sin^2 \theta} \cdot \sin \theta$

⑫: $S_{\text{梯} ABNQ} = \frac{1}{2} \cdot \frac{2p}{\sin^2 \theta} \cdot \frac{2p}{\sin^2 \theta} \cdot \sin \theta = \frac{2p^2}{\sin^3 \theta}$



⑬: 以 AB 为直径的圆与准线相切.

⑭: 以 AF, BF 为直径圆与 y 轴相切.



⑮: M, O, B 三点共线.
 N, O, A 三点共线.

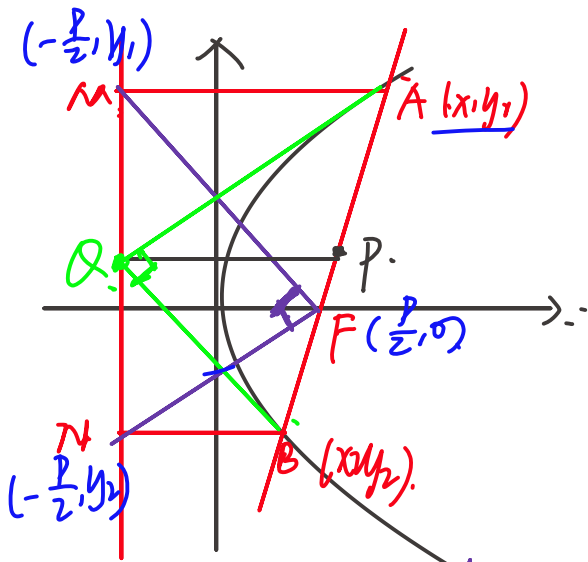
$$k_{OM} = \frac{y_1}{-\frac{p}{2}} = -\frac{2y_1}{p}$$

$$k_{OB} = \frac{y_2}{x_2} = \frac{2p}{y_2}$$

$$y_1 y_2 = -p^2$$

$$y_2 = \frac{-p^2}{y_1} \text{ 代入 } k_{OB} = \frac{2p y_1}{-p^2} = -\frac{2y_1}{p}$$

$$k_{OM} = k_{OB}$$

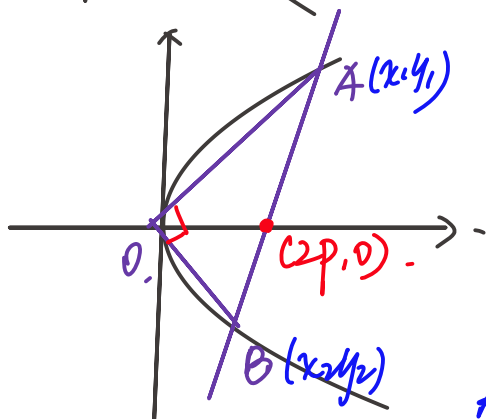


⑩ MF ⊥ NF AO ⊥ OB

$$k_{MF} = \frac{y_1}{-p}$$

$$k_{NF} = \frac{y_2}{-p}$$

$$k_{MF} \cdot k_{NF} = \frac{y_1 y_2}{p^2} = -1$$



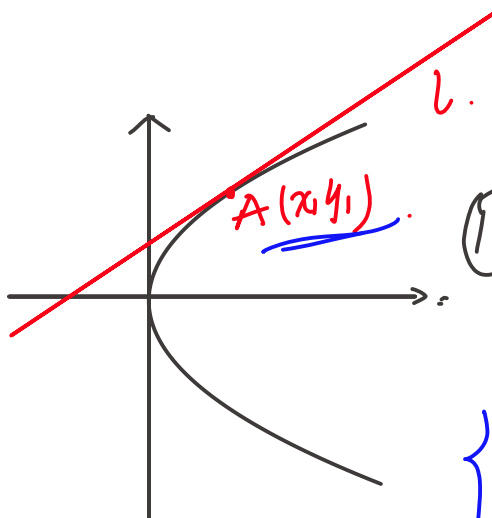
⑪ 已知 $OA \perp OB$ AB 过定点 $(2p, 0)$

$$\frac{x_1 x_2 + y_1 y_2}{2p} = 0$$

$$\frac{y_1^2}{2p} \cdot \frac{y_2^2}{2p} + y_1 y_2 = 0 \Rightarrow y_1 y_2 = -4p^2$$

$$AB \Rightarrow (y_1 + y_2)y = 2px + y_1 y_2$$

$$(y_1 + y_2)y = 2px - 4p^2 = 2p(x - 2p)$$



⑫: A(x1, y1) 处切线方程:

$$\underline{y_1 y = p(x + x_1)}$$

$$\left\{ \begin{array}{l} l: x = ky + b \\ y^2 = 2px \end{array} \right. \Rightarrow x_1 = ky_1 + b$$

$$y^2 = 2pk y + 2pb$$

$$y^2 - 2pk y - 2pb = 0$$

$$\Delta = 0$$

$$4p^2 k^2 + 8pb = 0$$

$$b = -\frac{pk^2}{2}$$

$$x_1 = ky_1 - \frac{pk^2}{2}$$

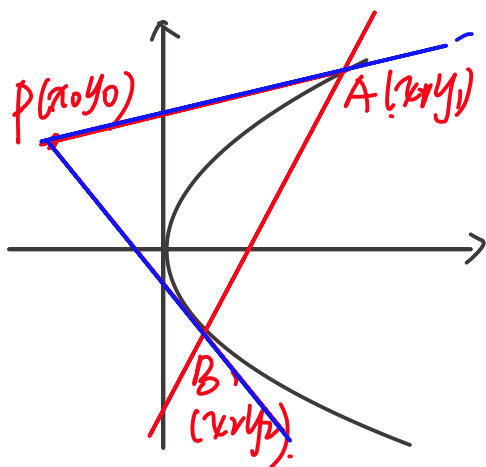
$$\frac{y_1^2}{2p} = ky_1 - \frac{pk^2}{2}$$

$$y_1^2 = 2pk y_1 - p^2 k^2 \Rightarrow (y_1 - kp)^2 = 0$$

$$k = \frac{y_1}{p}$$

$$x = \frac{y_1}{p}y - \frac{pk^2}{2} = \frac{y_1 y}{p} - \frac{p}{2} \cdot \frac{y_1^2}{p^2}$$

$$2px = 2y_1 y - y_1^2 \Rightarrow \underline{y_1 y = p(x + x_1)}$$



(19): AB: 切点弦方程: $y_0 y = p(x + x_0)$

三步走:

1. 推导 A(x_1, y_1) 处切线方程:

$$A: y_1 y = p(x + x_1)$$

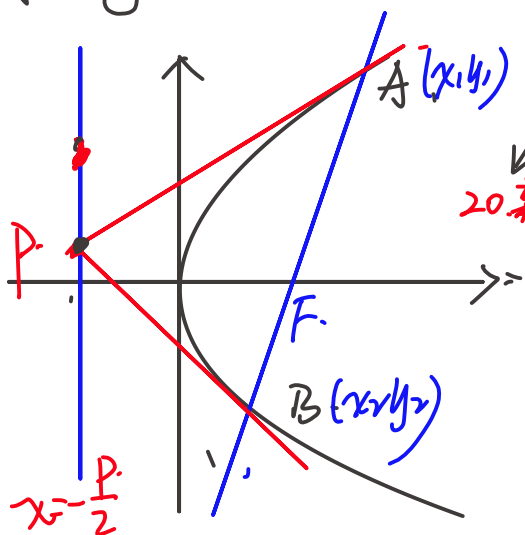
$$B: y_2 y = p(x + x_2)$$

2. P(x_0, y_0) 代入 $\begin{cases} y_0 y_1 = p(x_0 + x_1) \\ y_0 y_2 = p(x_0 + x_2) \end{cases}$

$$y_0 y_2 = p(x_0 + x_2)$$

3. 同构 AB: $y_0 y = p(x + x_0)$

$$\begin{cases} x^2 \rightarrow x_0 x \\ y^2 \rightarrow y_0 y \\ x \rightarrow \frac{x_0 + x}{2} \\ y \rightarrow \frac{y_0 + y}{2} \end{cases}$$



(20): P 是准线上一点, 过 P 作 PA, PB 与抛物线相切. 证: AB 过定点 $(\frac{p}{2}, 0)$

证: 推导 AB 切点弦方程. 2019

$$P(-\frac{p}{2}, t) \quad AB: ty = p(x - \frac{p}{2})$$

过 $(\frac{p}{2}, 0)$

(21): 过焦点 F 的直线交抛物线于 A, B

2023 新卷. 分别过 A, B 作切线: 证: 切线交点在定直线上

$$A: y_1 y = p(x + x_1)$$

$$B: y_2 y = p(x + x_2) \Rightarrow \frac{y_1}{y_2} = \frac{x + x_1}{x + x_2}$$

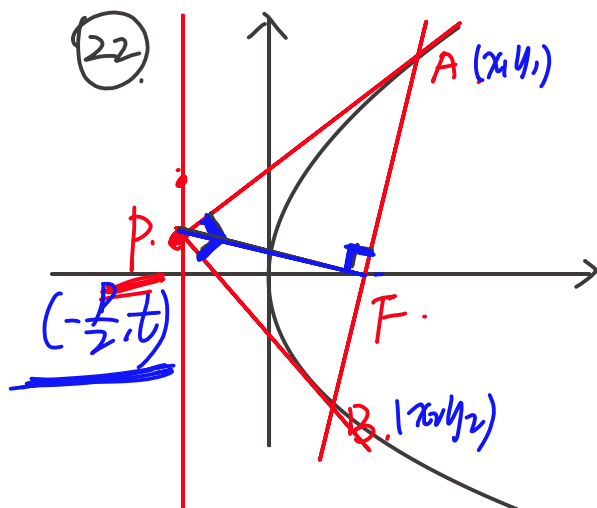
$$xy_2 + x_1y_2 = xy_1 + y_1x_2$$

$$(y_2 - y_1)x = y_1x_2 - y_2x_1$$

$$(y_2 - y_1)x = y_1 \cdot \frac{y_2^2}{2p} - y_2 \cdot \frac{y_1^2}{2p}$$

$$\cancel{(y_2 - y_1)}x = \frac{y_1y_2}{2p}(\cancel{y_2 - y_1})$$

$$x = \frac{y_1y_2}{2p} = \frac{-p^2}{2p} = -\frac{p}{2}$$



(22): $PA \perp PB$ · $PF \perp AB$

$$A: y_1y = px + px_1 \Rightarrow y = \frac{p}{y_1}x + p\frac{x_1}{y_1}$$

$$k_{PA} = \frac{p}{y_1}$$

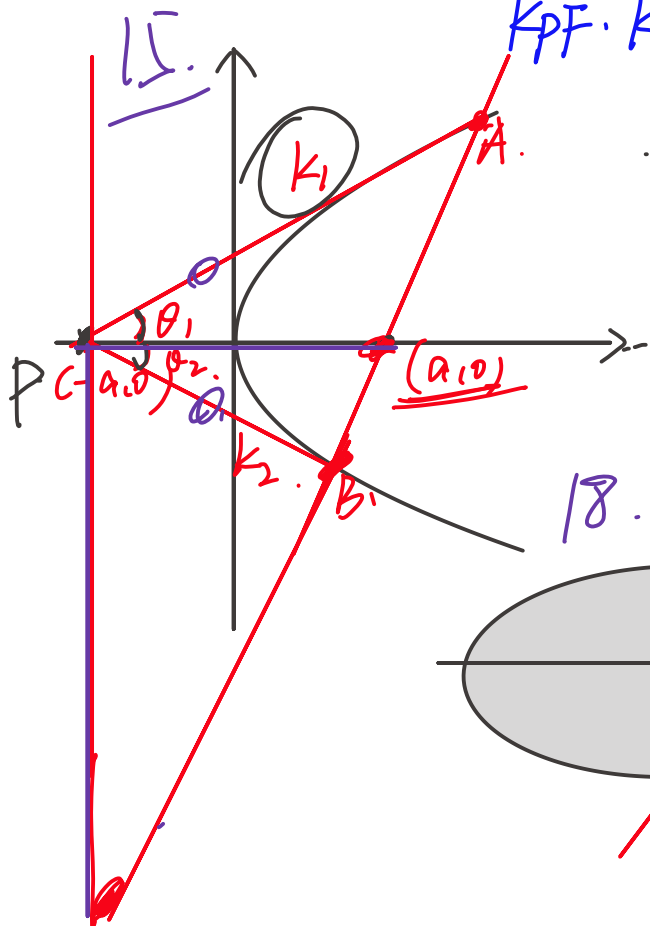
$$B: k_{PB} = \frac{p}{y_2}$$

$$k_{PA} \cdot k_{PB} = \frac{p^2}{y_1y_2} = \frac{p^2}{-p^2} = -1$$

$$k_{PF} = \frac{t}{p}$$

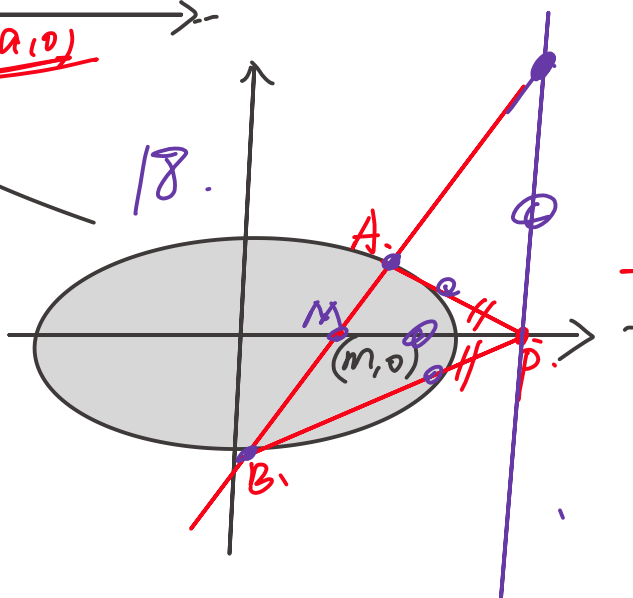
$$AB: ty = p(x - \frac{p}{2}) \Rightarrow k_{AB} = \frac{p}{t}$$

$$k_{PF} \cdot k_{AB} = -1$$

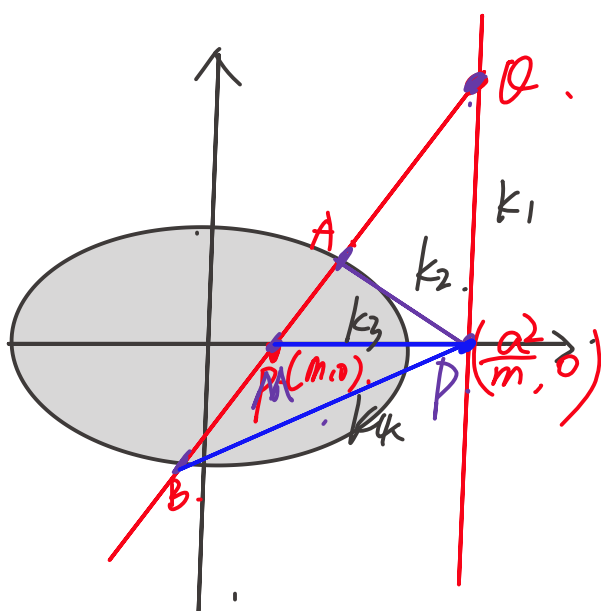


(23): AB 过 (a, 0). 在 x 轴上.

有定点 (-a, 0). 使得 $k_{AP} + k_{BP} = 0$.



定点 $P(\frac{a^2}{m}, 0)$
 $k_{AP} + k_{BP} = 0$.

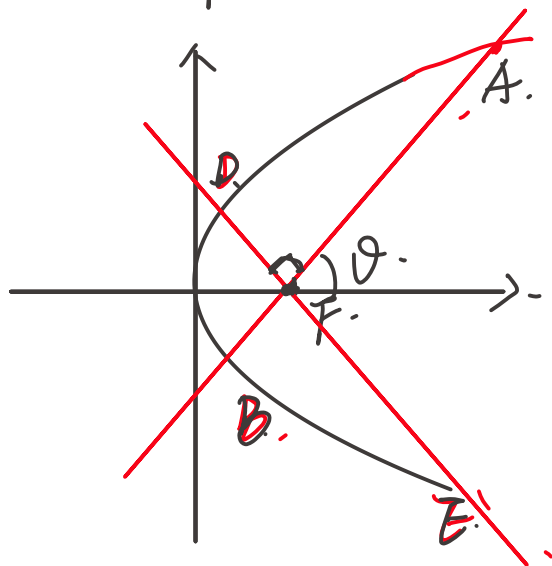


QAMB 调和关系.

PA, PB, PM, PB 调和关系.

$$k_1 = \infty, k_3 = 0,$$

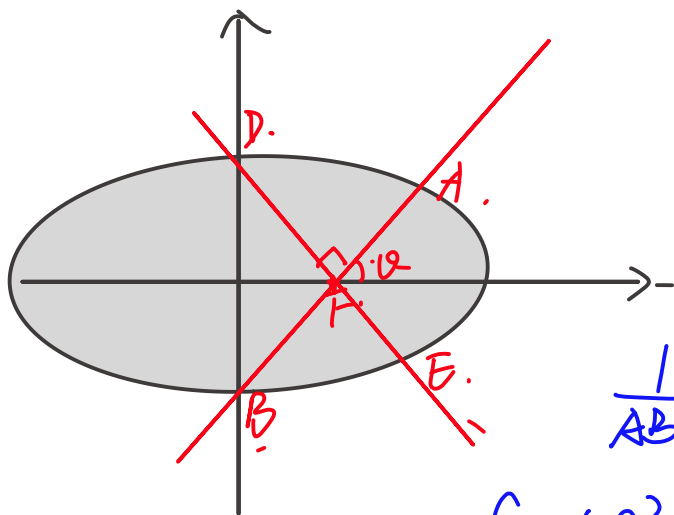
$$k_2 + k_4 = 0.$$



$$(24) \quad \frac{1}{AB} + \frac{1}{DE} = \frac{1}{2p} = \frac{2-e^2}{2p}$$

$$AB = \frac{2p}{\sin^2 \theta}$$

$$DE = \frac{2p}{\cos^2 \theta}$$



$$|AB| = \frac{\frac{2b^2}{a}}{1 - e^2 \cos^2 \theta}$$

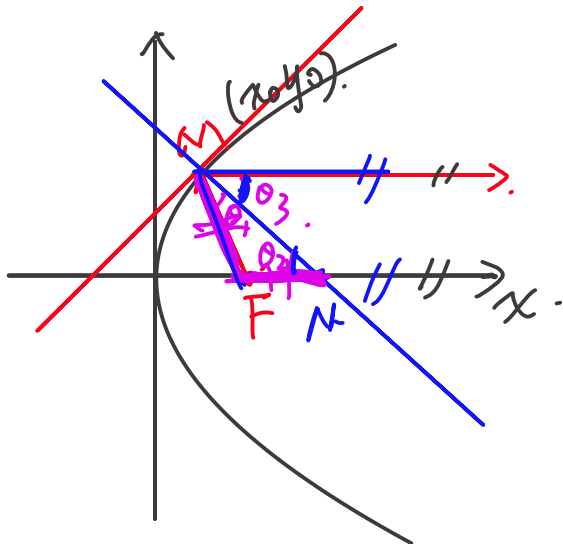
$$|DE| = \frac{\frac{2b^2}{a}}{1 - e^2 \sin^2 \theta}$$

$$\frac{1}{AB} + \frac{1}{DE} = \frac{2 - e^2}{\frac{2b^2}{a}}$$

$$2ep = 2 \cdot \frac{c}{a} \cdot \left(\frac{a^2}{c} - c \right) = 2 \frac{c}{a} \cdot \frac{b^2}{c} = \frac{2b^2}{a}$$

通径.

$$\frac{1}{AB} + \frac{1}{DE} = \frac{2 - e^2}{2ep}$$



②⑤ 圆锥曲线光学性质.

$M(x_0, y_0)$ 处切线: $y_0 y = p(x + x_0)$.

$$k_{MN} = -\frac{y_0}{p}$$

$$MN: y = -\frac{y_0}{p}(x - x_0) + y_0 \Rightarrow y = 0$$

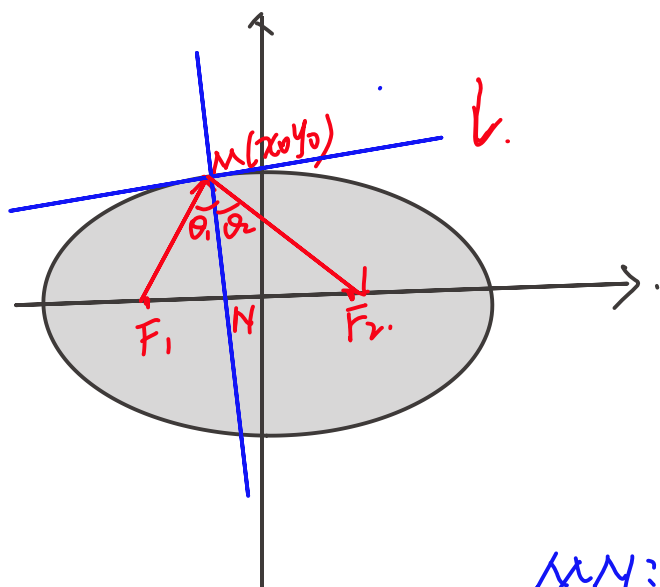
$$x_N = x_0 + p$$

$$|NF| = x_0 + p - \frac{p}{2} = x_0 + \frac{p}{2}$$

$$|MF| = x_0 + \frac{p}{2}$$

$$\theta_1 = \theta_2, \quad \theta_1 = \theta_3$$

$\theta_2 = \theta_3$, 平行于 x 轴



证: MN 是 $\angle F_1 M F_2$ 角平分线

$$l: \frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1 \quad \text{性质}$$

$$y = -\frac{b^2 x_0}{a^2 y_0} x + \frac{b^2}{y_0}$$

$$k_{MN} = \frac{a^2 y_0}{b^2 x_0}$$

$$MN: y = \frac{a^2 y_0}{b^2 x_0}(x - x_0) + y_0 \Rightarrow y = 0$$

$$x_N = x_0 - \frac{b^2 x_0}{a^2} = \frac{c^2}{a^2} x_0$$

$$NF_1 = \frac{c^2}{a^2} x_0 + c$$

$$NF_2 = c - \frac{c^2}{a^2} x_0$$

$$\frac{NF_1}{NF_2} = \frac{a + \frac{c}{a} x_0}{a - \frac{c}{a} x_0} = \frac{MF_1}{MF_2}$$

